

$$\xi_i^{(n)}$$

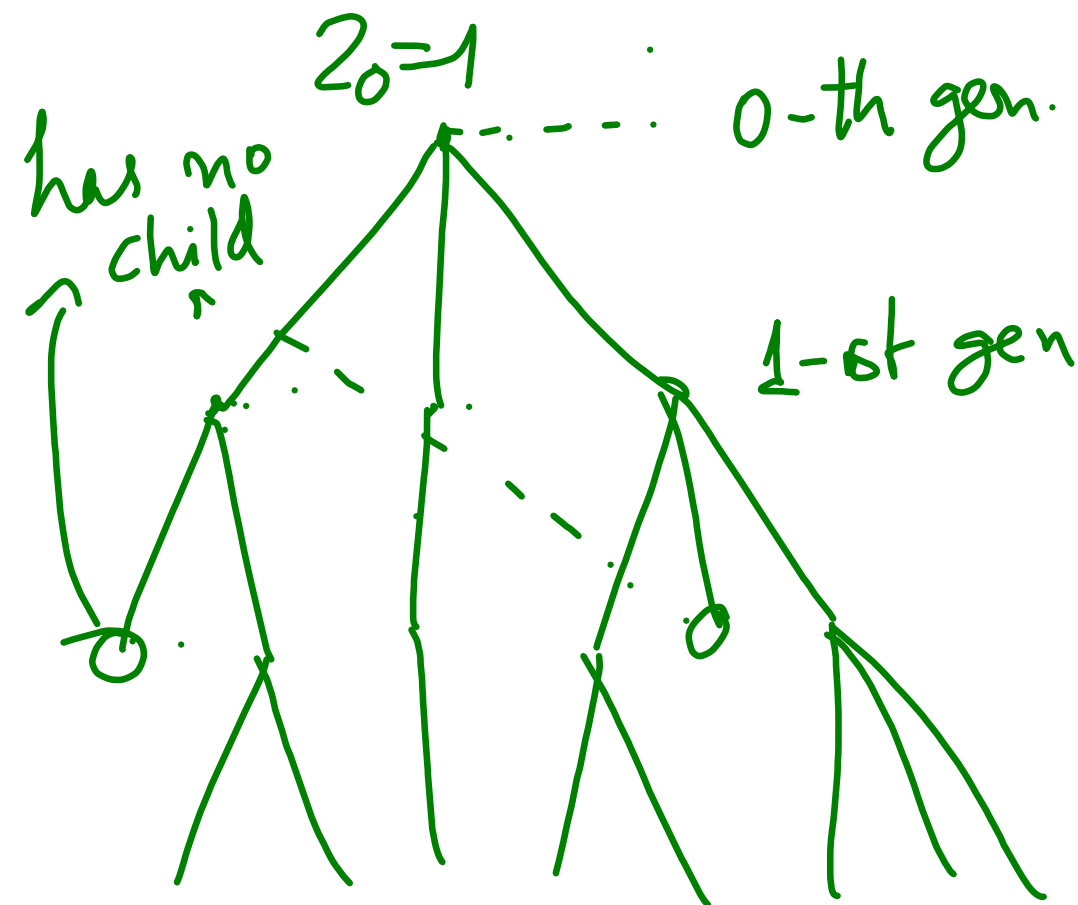
Branching Pr.

Galton Watson Pr.

Z_n = total size of the
nth gen.

$$P(\xi = k) = p_k, \quad k=0,1,\dots$$

↓
off spring distrⁿ.



At each gen, every
ind. generates offspring
acc. to the offspring
distrⁿ given by ξ .

$$\xi_1^{(0)} = 3$$

$$\xi_1^{(1)} = 2,$$

$$\xi_2^{(0)} = 1$$

$$\xi_3^{(1)} = 3$$

$$\xi_1^{(2)} = 0$$

Let $\xi_i^{(n)} : n \geq 0, i \geq 1$ denote the
no. of offsprings born to the i th
member of the n -th gen.

$$\{\xi_i^{(n)} : n \geq 0, i \geq 1\} \text{ iid.}$$

Define Z_n as the size of the n -th.

In this example, $Z_0 = 1, Z_1 = 3, Z_2 = 6,$
 $Z_3 = 8, \dots$

Note, $Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$ $\left(T = \text{time of extinction of the family} \right)$

Question : Does this family become extinct w.p. 1?

i.e, for some n , $Z_n = 0$.

$A = \{ Z_n = 0 \text{ for some } n \geq 1 \}$

$T = \inf \{ n \geq 1 : Z_n = 0 \}$. If $P(A) < 1$, then $P(T = \infty) > 0$

X_i gen. fn
R

N gen fn.

$$S_N = \sum_{i=1}^N X_i$$

$$P_{S_N}(s) = Q(R(s))$$

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n)}$$

Let G be the pgf
of ξ , i.e., $G(s) = \sum_{k=0}^{\infty} p_k s^k$

$$P_{n+1}(s) = P_n(G(s))$$

$$E(Z_{n+1}) = E(Z_n) E(\xi) = E(Z_{n-1}) (E(\xi))^2 = \dots = E(Z_0) (E(\xi))^{n+1}$$

Let P_n be the gen
fn. of Z_n .

$$P_n(s) = \sum_{k=0}^{\infty} P(Z_n = k) s^k$$

$$\forall n \geq 0.$$

Let $m = E(\xi)$.

Then, $E(Z_{n+1}) = m$

Assume $V(\xi) = \sigma^2$. Compute $\text{Var}(Z_n)$.

In gen., expression of P_n is difficult to obtain.

$$P_{n+1}(s) = P_n(\underbrace{G(s)}_x) = P_{n-1}(G(\underbrace{G(s)}_x))$$

$$\begin{aligned} P_0(s) &= \sum_{k=0}^{\infty} P(Z_0=k) s^k \\ &= 1 \times s = s \end{aligned}$$

$$\begin{aligned} &= P_{n-2}(G(G(G(s)))) \\ &\vdots \\ &= P_0(\underbrace{G(G(\dots G(s) \dots))}_{(n+1)\text{-fold comp.}}) \\ &= \underbrace{G(G(\dots G(s) \dots))}_{(n+1)\text{-fold con.}} \end{aligned}$$

For any $n \geq 1$,

$$P_n(s) = \underbrace{G(G(\dots G(s) \dots))}_{n\text{-fold conv.}}$$

$$= G(P_{n-1}(s))$$

$$P_n(s) = G(P_{n-1}(s))$$

$\forall n \geq 1$.

B_n events
 $B_n \subseteq B_{n+1}$
 $P(\bigcup_n B_n)$
 $\geq \lim_{n \rightarrow \infty} P(B_n)$

$$A = \{Z_n = 0 \text{ for some } n \geq 1\}$$

$$= \bigcup_{n=1}^{\infty} A_n$$

$$A_n \subseteq A_{n+1}$$

$$P = P(A) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = \lim_{n \rightarrow \infty} P(A_n)$$

$$P = \lim_{n \rightarrow \infty} P_n(0)$$

$$= \lim_{n \rightarrow \infty} P_n(0)$$

$$\begin{aligned}
 A_n &= \{Z_n = 0\} \\
 P(A_n) &= P(Z_n = 0) \\
 &= P_n(0)
 \end{aligned}$$

Note $P_{n+1}(0) = G(P_n(0))$

Taking limit as $n \rightarrow \infty$,

$$p = G(p).$$

Thus, p is a fixed pt.
of G .

$$x_{n+1} = G(x_n)$$

$$x_n \rightarrow x$$

$$x = G(x)$$

Assume $p_1 \neq 1$.

⊗

$G'(s) \geq 0$, on $[0, 1]$, i.e.

G is non-dec.

Let τ be any fixed pt
of G in $[0, 1]$, i.e., $G(\tau) = \tau$.

$\tau \geq 0, \Rightarrow G(\tau) \geq G(0)$ $\tau \in [0, 1]$.
 $\Rightarrow \tau \geq G(0)$.

$p_1 \neq 1$

$2n \geq 1$

$\forall n \geq 1$

$2n \rightarrow 1$.

$$G(T) \geq G(G(o))$$

$$T'' \geq G(G(o))$$

By induction,

$$T \geq \underbrace{G(G(\dots G(o)\dots))}_{n\text{-fold comp. } \forall n \geq 1}$$

Taking limit, $= P_n(o)$

$$T \geq \lim_{n \rightarrow \infty} P_n(o) = p.$$

Thm :- $\textcircled{x}$ ρ is given by the
smallest fixed pt. of G in $[0,1]$.

$\frac{Z_n}{\mu^n} | Z_n \rightarrow 0$. G has 1 fixed pt, namely 1, ρ .

if $G'(1) = m \leq 1$.

G has two fixed pts $\rho, 1$ if
 $m = G'(1) > 1$.

· Start with a single dice.

· Roll the dice. If it shows,

1, 2 $\rightarrow$ the dice is taken back.

3 $\rightarrow$ the dice is given back,

4, 5 $\rightarrow$ the dice is given back
& another dice is also given.

6 $\rightarrow$ dice is given., two more